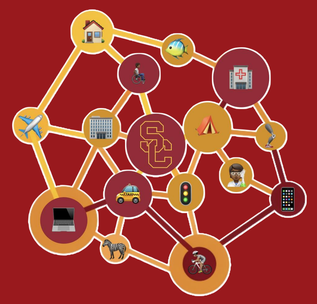
Network Reconnaissance



Assuage: Improving mHealth Accessibility and Adherence for Remote Patient Monitoring

Luis Millan, Alyssa Donawa, Corey E. Baker
Ming Hsieh Department of Electrical and Computer Engineering
University of Southern California

USC Viterbi School of Engineering

Introduction

Problem

- The widespread use of mobile devices has made mHealth systems appealing for reaching disadvantaged populations[4,5]
- Remote patient monitoring (RPM) through mobile apps is well-received by patients, including underserved populations, and has the potential to improve health outcomes[5]
- RPM can include objective data via sensor devices or subjective data via patient-reported outcomes (PROs)
- **Nevertheless, determining how to design and deliver mHealth full-stack systems and apps (server and client side) for vulnerable populations needs further investigation**[6]

Approach

- The **NetRecon Lab**[2] developed **Assuage**[1], a **HIPAA-compliant mobile iOS application** designed to facilitate RPM
- Build upon an external framework, CareKitEssentials, to enhance data logging capabilities, allowing users to efficiently log distress levels
- Developed methods for calculating task progress using median and average values, providing accurate and meaningful insights
- Incorporated comprehensive testing protocols, including the creation of test cases for various scenarios and the use of mocking frameworks to simulate real-world data

Open-Source Frameworks

Apple

HealthKit:

- Provides a central repository for health and fitness data on iPhone and Apple Watch
- Allows apps that provide health and fitness services to share their data, creating a comprehensive and personalized health overview for users

ResearchKit:

- Facilitates the collection of data on a large scale, aiding researchers in gathering robust and meaningful data[3]

CareKit:

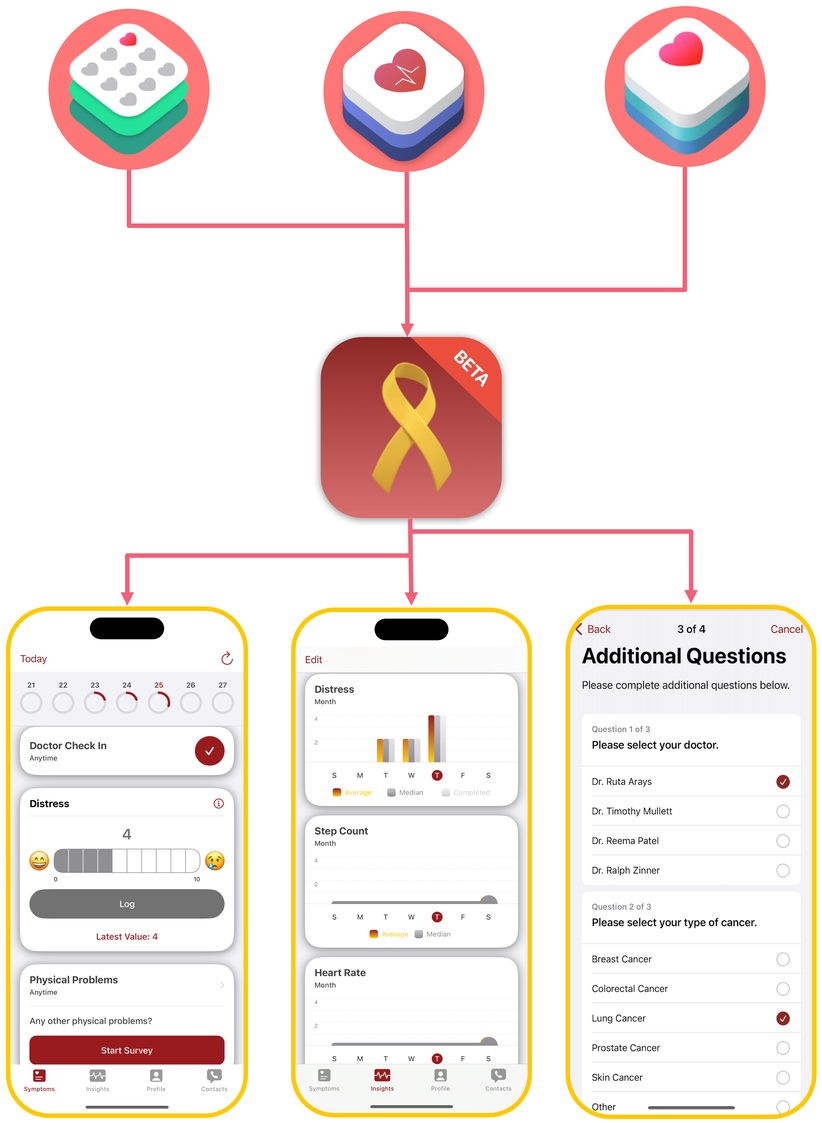
- Comprising of three packages: CareKit, CareKitUI, and CareKitStore
- Provides view controllers, enabling developers to create intuitive and user-friendly health apps[3]

NetRecon Lab

CareKitEssentials:

- Provides essential cards, views, models, protocols, and extensions to expedite building CareKit-based applications
- Offers a variety of tools that streamline the development process

Assuage



- This framework expedites the creation of CareKit-based applications. It provides essential components, reducing the learning curve and making development more efficient
- Ideal for situations with limited resources or time, it enhances product quality by giving developers time to focus on the unique aspects of their application

Findings

- Enhanced our data processing by adding a flexible feature to our configuration tool
- Allows us to track and analyze different types of health data better, which is crucial for providing accurate and personalized insights
- Introduced a new option to specify the data type being viewed, making it easier to filter and understand the information presented
- Significantly increased our code coverage by adding new test cases, covering various scenarios and edge cases
- This thorough testing process helps ensure that our app functions reliably and meets high-quality standards

Summary

Problem

- Despite the benefits of mHealth systems and RPM, there are difficulties in providing adequate health services to vulnerable populations
- More research is needed to optimize the design and delivery of mHealth applications, focusing on both server and client-side solutions
- Ensuring that mHealth technologies are user-friendly and accessible to all populations remains a critical issue

Findings

- Implemented flexible feature to the configuration tool, accurately tracking and analyzing diverse health data types for personalized insights
- Introduced a new option to specify the data being viewed, enhancing the ability to filter and visualize data
- After implementing new features, expanded test cases to cover various scenarios and edge cases, ensuring reliability and high-quality standards

Future Work

- Develop strategies to enhance motivation and engagement to improve patient adherence
- Determine the ideal frequency and methods for prompting PROs to maximize engagement without causing fatigue

Acknowledgements

I would like to thank Dr. Corey Baker (NetRecon Lab) for providing me the opportunity to work on this project, as well as PhD candidate Alyssa Donawa for her guidance. Thank you as well to the Amazon-Sure program for providing this experience.

References

1. Alyssa Donawa Assessing Cancer Patient Usability of a Mobile Distress Screening App COMPASS '22: ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies 29 June 2022- 1 July 2022 Seattle WA USA
2. Network Reconnaissance Lab, University of Southern California - https://netrecon.usc.edu/baker/research
3. Apple, ResearchKit and CareKit - https://www.researchandcare.org
4. Lindsay Satterwhite Mayberry, Courtney R. Lyles, Brian Oldenburg, Chandra Y. Osborn, Makenzie Parks, Monica E. Peek. mHealth Interventions for Disadvantaged and Vulnerable People with Type 2 Diabetes. Paper presented at: Current Diabetes Reports; Published online: November 25, 20191. Accessed: July 27, 2024.
5. Will L. Tarver, David A. Haggstrom. The Use of Cancer-Specific Patient-Centered Technologies Among Underserved Populations in the United States: Systematic Review. Paper presented at: Journal of Medical Internet Research; Published online: April 23, 20191. Accessed: July 27, 2024.
6. Signe Hanghøj, Kirsten A. Boisen, Maiken Hjerming, Abbey Elsbernd, Helle Pappot. Usability of a Mobile Phone App Aimed at Adolescents and Young Adults During and After Cancer Treatment: Qualitative Study. Paper presented at: JMIR Cancer; Published online: January 2, 20201. Accessed: July 27, 2024.

USC University of Southern California

amazon

Install Assuage if you have **TestFlight** *(iOS only)*

Email: lm_879@usc.edu